

Observable Classification Theory for SOFs

Wall Records, Observable Invariants, Taxonomy, and Smooth Local Models

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This paper is Part XI of the RIME program. Paper VII isolates generic accessibility completion; Paper VIII fixes the static SOF object layer; Paper IX studies observable dynamics over SOFs; and Paper X isolates the Universal Observable Pipeline together with a five-layer SOF Registry. Paper XI turns that registry into an observable classification layer: given registered SOFs, what feature space classifies their observable behavior?

Abstract

Problem. The SOF Registry records species, SOF objects, observable ladders, dynamics, and diagnostics. This makes cross-species comparison possible, but it does not yet say what type a registered SOF has at the observable level. The next question is not whether every wall belongs to a known singularity class. The question is which observable signatures and invariants classify the behavior of registered SOFs.

Approach. We introduce a four-layer organization. First, **wall records** collect where observable changes occur: rank jumps, support jumps, accessibility jumps, plateau intervals, collision loci, terminal boundaries, and related discriminants. Second, **wall signatures** upgrade a wall from an event to an observable signature: rank type, support type, repair type, oscillation type, plateau type, or equivalently changes such as $(\Delta R_1, \Delta R_2, \Delta D, \Delta \tau, \Delta P_d)$. Third, the ordered sequence of signatures along a deformation is an **observable wall record**, and the multiset of such signatures is the **wall spectrum** $\text{Spec}_W(\mathcal{F})$. Fourth, the **Observable Wall Taxonomy** groups wall spectra into collision, repair, terminal-side, plateau/rate, nonsmooth/discrete, and bridge/incidence wall families. Smooth ADE models enter only afterward as candidate local normal forms for sufficiently smooth discriminant branches.

Results. The main result is an observable classification schema, not a complete classification theorem. It turns wall records into computable features such as crossing count, wall-cell count, repair index, oscillation index, depth index, rate ratio, spectral gap, and codimension. The schema is supported by cross-species and spectral audits: Rubik spectral snapshots, a Rubik spectral-deformation audit with 16 pairwise A_1 -type collision candidates and simultaneous pair-gap response diagnostics, a 2D commutativity-discriminant slice with one hit on the chosen 20×20 evaluation grid, activation-wall diagnostics, quantum accessibility contrasts, Markov frozen-pair diagnostics, graph Laplacian spectral-gap sensitivity, a barrier-option stopping-sector diagnostic, and a transformer activation-sector diagnostic. The audits show that registered species expose distinct wall types and that several important cases are discrete, piecewise-smooth, degenerate, stochastic, activation-based, or mechanism-specific rather than smooth ADE-type singularities. A separate 24-record coverage census contains 15 initially eligible records. Classes B–F meet the provisional three-record, two-species, two-deformation curation target; Class A remains a single-species Rubik gap. An observable-status trajectory audit records 190 sampled pair-status changes for a three-step GridWorld obstacle path, 4 changes at the isolated SIR $\beta = 0$ boundary, and 6 changes at the terminal point of a graph edge-weight path. A definition-compatible redundancy audit over 166 configurations finds that 3 empirical principal components explain both 90% and 95% of snapshot variance for the tested ensemble; this PCA summary is ensemble-composition dependent and does not produce a six-invariant

basis theorem. Finally, a block-restricted eigenbranch-continuation audit finds no $A_2 \rightarrow A_1 + A_1$ split candidate on the tested Rubik slices.

Implications. Paper XI narrows the classification problem and supplies the feature-extraction layer for later diagnostics. ADE-type local models may be useful for spectral collision or smooth discriminant maps, but they cannot be the default language for all SOF walls. RIME at this stage controls wall location and observable diagnostics; it does not yet provide general wall-crossing transformation laws. A full taxonomy must first distinguish wall records, wall signatures, taxonomy classes, and smooth local models.

Notation Table

Symbol	Meaning
$\mathcal{F} = (V, \{Q_i\}, \mathcal{X})$	Sectorized Observable Framework
$\mathcal{F}_t$	SOF deformation
$O(t)$	observable trajectory or shadow
Σ_O	wall or discriminant locus of O
Σ_{comm}	commutativity locus in generator-weight moduli
Δ_O	target discriminant in observable space
$\mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma)$	wall record for a SOF object, observable ladder, and deformation geometry
W_i	one observable wall signature event in a deformation record
$\text{Sig}_W(\mathcal{F})$	computable wall signature
$\text{Spec}_W(\mathcal{F})$	multiset of wall signatures
WallAssign	record-producing assignment $(\mathcal{F}, \mathcal{L}, \Gamma) \mapsto \mathcal{W}$
$N_{\text{cross}}, N_{\text{cell}}$	crossing count and wall-cell count
R_W, p_W, O_W, P_W	repair, persistence, oscillation, and plateau indices
$\bar{D}, \rho_W, \gamma_W, c_W, \delta_W$	depth, rate-ratio, spectral-gap, codimension, and density diagnostics
R_1, R_2, D	accessibility observable ladder
$\tau(O)$	characteristic time scale when defined
$P_d(t)$	plateau function
A_k, D_k, E_k	Arnold ADE singularity families

1 Introduction

The post-Paper-VII SOF arc has three layers:

Paper	Role
Paper VIII	SOF object layer
Paper IX	observable trajectories over SOFs
Paper X	Universal Observable Pipeline and five-layer SOF Registry

Paper XI asks what comes after the registry.

The registry is a taxonomy input:

species $\rightarrow$ SOF object $\rightarrow$ observable ladder $\rightarrow$ dynamics $\rightarrow$ diagnostics.

Paper X records which SOFs exist and how they enter the observable pipeline. Paper XI asks what observable type those SOFs have. In other words, the registry is a list of realized species, while Paper XI extracts the observable features by which those species can be compared.

Paper XI asks whether registered wall records can be grouped into classes. The answer is not yet a complete classification. The safe claim is that the classification problem splits into observable signatures, wall spectra, invariants, normal forms, and smooth local models.

This paper is deliberately conservative. It does not claim that Arnold ADE classification applies to all SOF walls. It treats ADE as one candidate local model for sufficiently smooth discriminant-like wall records and records where that language does not apply. The organizing object is the wall record, not an ADE class.

The RIME papers up to this point mostly control **wall location**: where support, rank, accessibility, plateau, or collision behavior changes. They do not yet give a general wall-crossing law for how all invariants transform across every wall. Paper XI therefore asks a prior organizational question: what kinds of observable walls have been registered, and how should they be organized before any transformation law is attempted?

The answer has four layers:

Layer	Object	Question
Wall Records	loci and observable changes	Where do walls occur?
Wall Signatures	computable invariants of records	What does the wall look like?
Observable Wall Taxonomy	classes of signatures	What type of wall is it?
Smooth Wall Models	local smooth-discriminant models	When can ADE-like tools apply?

For later diagnostic reports, the same layer becomes a feature extractor. An application-facing SOF report need not use theorem proofs directly. It can emit quantities such as crossing count, wall-cell count, repair index, plateau or oscillation index, and depth index, then interpret them as observable diagnostics.

This also extends the categorical architecture begun in Paper VIII:

$$\begin{aligned} \text{Source} &\rightarrow \text{SOF}_{\text{str}} \rightarrow \text{Observable ladder} \rightarrow \text{Observable dynamics} \\ &\rightarrow \text{Wall records} \rightarrow \text{Observable invariants} \rightarrow \text{Wall taxonomy.} \end{aligned}$$

Paper XI does not prove that this last passage is a categorical invariant of SOFs. It introduces the record-producing layer needed to make the passage explicit once the observable ladder and deformation geometry have been chosen.

2 Wall Records

2.1 Definition

A **wall record** for a SOF deformation is the data

$$\mathcal{W}(\mathcal{F}_t, O) = (\mathcal{F}_t, O(t), \Sigma_O, \Delta_O, \mathcal{D}_O),$$

where:

1. $\mathcal{F}_t$ is a specified SOF deformation;
2. $O(t)$ is an observable shadow or continuous observable field;
3. Σ_O is the locus where the observable changes qualitative type;
4. Δ_O is the target discriminant when a pullback description exists;
5. $\mathcal{D}_O$ is the measured diagnostic package, such as $\tau(O)$, repair, plateau, frozen-pair count, or spectral-gap change.

The wall record is the object that Paper XI attempts to classify.

2.2 Wall Assignment

Paper VIII defines strict SOF morphisms and the category SOF_{str} . Paper XI adds a record-producing construction

$$\text{WallAssign} : (\mathcal{F}, \mathcal{L}, \Gamma) \mapsto \mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma),$$

read as: a realized SOF $\mathcal{F}$, together with a chosen observable ladder $\mathcal{L}$ and deformation geometry or sampled path Γ , determines the wall records observed by that ladder along that geometry.

This notation is intentionally modest. It is a bookkeeping assignment, not a claim that wall records form a categorical morphism out of SOF_{str} . The present paper does not prove invariance under every weak SOF comparison. It only asserts that wall records are naturally recorded after SOF realization, observable extraction, and deformation choice.

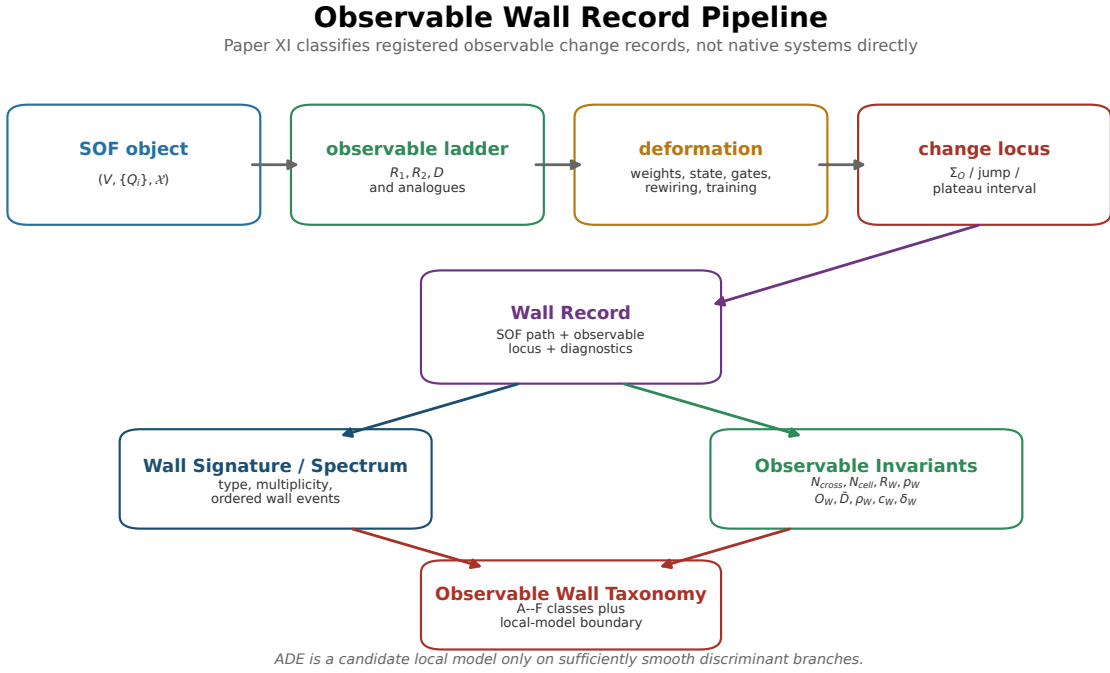


Figure 1. Observable wall-record pipeline. A realized SOF, observable ladder, deformation geometry, and change locus produce a wall record. Wall signatures, wall spectra, and computable invariants then feed the Observable Wall Taxonomy. ADE is only a candidate local model on sufficiently smooth discriminant branches.

2.3 Aggregate Wall Record

For a realized SOF deformation, the aggregate wall record is the collection of observable change loci that have been registered:

$$\mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma) = \{\text{rank jumps, support jumps, accessibility jumps, plateau intervals, collision loci, ...}\}.$$

For a one-parameter deformation, the observable wall record is more usefully viewed as an ordered sequence

$$\mathcal{W}_{\text{obs}}(\mathcal{F}_t) = (W_1, W_2, \dots, W_k),$$

where each W_i is an observable wall signature event encountered along the path. This sequence is analogous in spirit to a critical sequence in Morse theory, a bifurcation diagram, a persistence barcode, or an RG-flow history, but it is not identified with any of these objects. It is the SOF-native record of observable changes along a chosen deformation.

The ellipsis is important: the registry is open. A new wall type may enter only after its SOF object, observable ladder, deformation geometry, jump or discriminant locus, and diagnostic package are explicit.

2.4 What Is Not Being Classified

Paper XI does not classify native systems directly. Rubik, quantum gates, Markov chains, graphs, neural networks, and Yang-like systems are not compared through their native coordinates.

The classification target is the wall record after SOF realization.

3 Wall Signatures

Wall records say where an observable changes. A **wall signature** records what the wall looks like in computable terms.

The basic object is an observable signature event

$$W_i = (\text{rank type, support type, repair type, oscillation type, plateau type}).$$

For accessibility and trajectory diagnostics, the same information can be recorded as an observable-difference vector

$$W_i = (\Delta R_1, \Delta R_2, \Delta D, \Delta \tau, \Delta P_d).$$

The first notation records qualitative type; the second records measured observable change. Paper XI classifies these observable signatures, not bare wall events.

A typical signature has the form

$$\text{Sig}_W(\mathcal{F}_t, O) = (c, k, p, m, r, b, \dots),$$

where:

1. c records the number of detected components or cells;
2. k records codimension when a meaningful ambient geometry is known;
3. p records persistence: persistent, transient, terminal, or degenerate;
4. m records temporal profile: monotone, oscillatory, plateau, flat, or discontinuous;
5. r records repair status: repairable, unrepaired, terminal, or not applicable;
6. b records boundary type: collision, rank/support jump, absorbing boundary, activation kink, stopping boundary, or filtration degeneration.

This is not yet a complete invariant. It is a computable signature schema for comparing wall records without pretending that all walls belong to one smooth classification theory.

3.1 Wall Spectrum

The **wall spectrum** of a realized SOF deformation is the multiset of wall signatures observed across the chosen observable ladder:

$$\text{Spec}_W(\mathcal{F}_t) = \{\{\text{Sig}_W(\mathcal{F}_t, O_a) : O_a \in \mathcal{L}(\mathcal{F}_t)\}\},$$

where $\mathcal{L}(\mathcal{F}_t)$ denotes the observable ladder under audit. Equivalently, one may write schematically

$$\text{Spec}_W(\mathcal{F}_t) = \{\{\text{rank wall, accessibility wall, plateau wall, ...}\}\}.$$

The multiset notation allows repeated wall types. A species may exhibit two distinct repair walls, or both a collision wall and a plateau wall, and these should not be collapsed into a single label.

3.2 Observable Wall Invariants

Observable signatures produce numerical invariants. These invariants are not claimed to be complete; they are the first stable coordinates for comparing wall spectra and for passing diagnostic summaries to later SOF reports.

Invariant	Definition	Example audited value
Crossing count N_{cross}	number of sampled wall crossings or adjacent observable-status changes along a declared deformation path	GridWorld obstacle path: 190 sampled pair-status changes
Wall-cell count N_{cell}	number of sampled cells or points on an evaluation grid satisfying a declared wall predicate	Rubik 2D slice: one Σ_{comm} hit on the chosen 20×20 evaluation grid
Repair index R_W	$D_{\text{repaired}}/\text{frozen}_{R_1}$ when both are defined	Clifford+CNOT: $6/6 = 1.0$; Pauli: $0/8 = 0$
Repair persistence p_W	parameter-range length, measured relative to the normalized deformation parameter, for which a repaired channel remains accessible after activation	CNOT-strength interpolation: threshold 0.55, $p_W = 0.45$, stability 100%
Oscillation index O_W	number of sign changes or oscillatory reversals in a plateau diagnostic	Rubik generator-weight plateau: 3; Yang-like state mixing: 0
Plateau index P_W	number, length, or collapse profile of plateau intervals in $P_d(t)$	Yang-like and training-coupled diagnostics supply initial plateau profiles
Depth index $\bar{D}$	maximum recorded D over sector pairs, with frozen entries marked separately	graph/Rubik-style frozen entries may use 999 as a sentinel; transformer diagnostic has $\bar{D} = 2$
Rate ratio ρ_W	slow observable time divided by fast observable time	NN GD+WD proxy: about $2\times$; ridge row/null diagnostic: about $68553\times$
Spectral gap γ_W	normalized second-gap or transition-gap diagnostic for a registered species	Clifford+CNOT registry example: about 0.73; Pauli control: 0
Codimension c_W	local or computational codimension when an ambient geometry is available	Σ_{comm} : 11; A_1 : 1; A_2 : 2
Wall density δ_W	fraction of species in the Paper XI wall-density taxonomy sample exhibiting a given wall type	15-entry sample shown below

These quantities make the taxonomy operational. A later SOF diagnostic report can emit a wall spectrum together with N_{cross} , N_{cell} , R_W , P_W , O_W , $\bar{D}$, ρ_W , γ_W , c_W , p_W , and δ_W whenever the corresponding observable layer is defined. For example, a transformer diagnostic need

not prove a wall theorem; it may report crossing count, wall-cell count, repair index, plateau or oscillation index, and depth index as observable features of the model under the chosen sectorization.

3.3 Wall-Density Taxonomy Sample

In the 15-entry Paper XI wall-density taxonomy sample, wall density is computed as

$$\delta_W(\text{type}) = \frac{\#\{\text{taxonomy-sample species carrying that wall type}\}}{15}.$$

The table below is generated by the wall-density support artifact listed in Appendix C. The resulting initial density table is:

Wall type	Density	Dominant source
B repair	33.3%	5 species; the most common wall type
F bridge/incidence	20.0%	Rubik naturally carries bridge and incidence walls
C terminal-side	13.3%	barrier option and absorbing Markov examples
D plateau/rate	13.3%	Xu ridge and mechanism-separated dynamics
E nonsmooth/discrete	13.3%	graph rewiring and ReLU kink diagnostics
A spectral collision	6.7%	Rubik spectral collision branch; rarest in this sample
no registered wall type	20.0%	3 of 15 entries are static or structural species

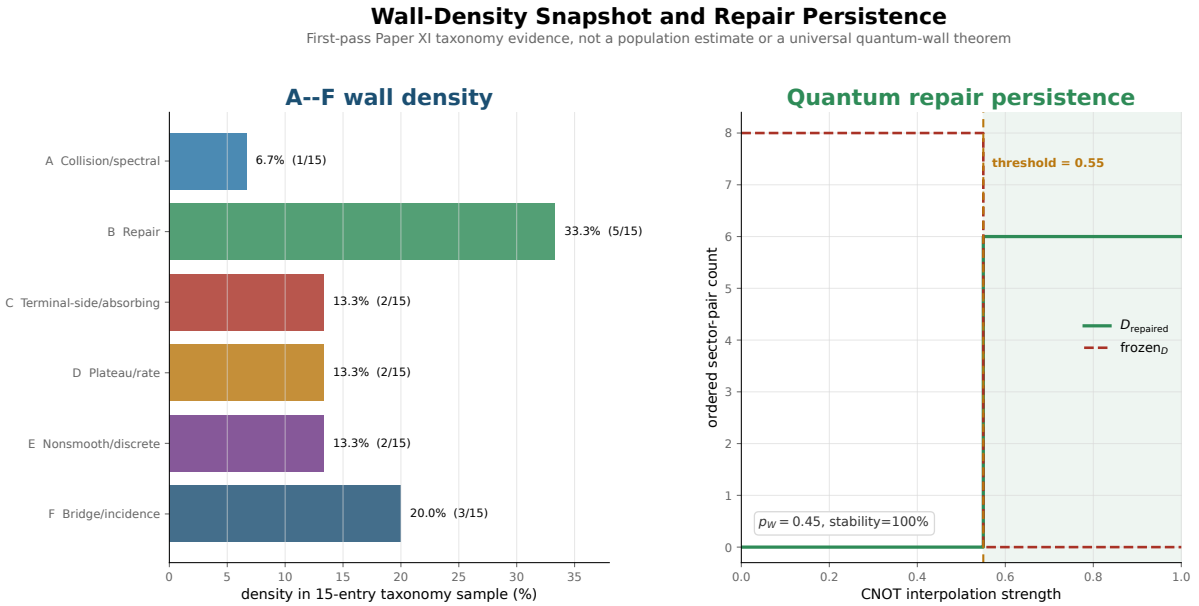


Figure 2. Wall-density and repair-persistence evidence. The left panel reports A–F wall densities in the 15-entry Paper XI taxonomy sample, which is distinct from the frozen 16-entry Paper X Registry release snapshot. The right panel shows the CNOT-strength repair threshold at 0.55, persistence $p_W = 0.45$, and 100% post-activation stability on the tested grid.

The snapshot is not a theorem and should not be read as a population estimate for all possible SOFs. It is a taxonomy-sample statistic. Its immediate interpretation is useful: repair walls are the most common in the taxonomy sample, echoing the Paper VII focus on generic completion and repair; spectral collision walls are the rarest, suggesting that Paper XI requires more continuous deformation models before spectral-collision taxonomy can be considered representative.

3.4 Wall-Record Coverage Census

Species prevalence alone does not measure whether each taxonomy class has enough independent wall records. We therefore maintain a second computational census whose unit is a wall record, not a species. A record may carry multiple class labels when one deformation has several observable features, but repeated records do not create new species in the prevalence denominator.

For this initial audit, a record is **eligible** when it has an explicit deformation, an explicit change locus, a measured signature, and an existing evidence file. The provisional coverage target for each class is at least three eligible records, two distinct species, and two deformation origins. This is a curation threshold, not a classification theorem criterion.

The census contains 24 records, 32 class memberships, and 15 eligible records:

Class	Registered records	Eligible records	Eligible species	Deformation	
				ori- gins	Coverage
A collision/spectral	3	2	1	2	gap
B repair	9	5	5	5	pass
C terminal-side/absorbing	4	3	3	3	pass
D plateau/rate	4	4	4	4	pass
E nonsmooth/discrete	4	3	3	3	pass
F bridge/incidence	8	3	3	3	pass

The result isolates the remaining sampling problem: Class A has multiple Rubik records and two deformation origins, but still only one eligible species. Additional non-Rubik continuous spectral-collision models are needed. The other five classes meet the provisional record/species/deformation target, but this does not make the six-class taxonomy complete or statistically representative. Full generated tables and record provenance are stored in the repository artifacts listed in Appendix C.

3.5 Definition-Compatible Redundancy Audit

The ten wall invariants listed above do not share one sampling unit. Codimension requires an ambient geometric model, δ_W is a cross-species statistic, and O_W , P_W , and p_W require trajectories. They therefore must not be inserted into one snapshot correlation matrix by replacing them with unrelated proxies.

The definition-compatible redundancy audit instead computes eight snapshot diagnostics on the same 166 controlled configurations: direct-frozen fraction, Lie-terminal fraction, repair index, word-bridge only fraction, Lie-bridge-only fraction, mean and maximum word depth, and log observable norm ratio. The strongest empirical correlations are:

Pair	Correlation r
direct-frozen fraction / mean word depth	+0.932
mean / maximum word depth	+0.938
direct-frozen fraction / maximum word depth	+0.923
direct-frozen / Lie-terminal fraction	+0.886
Lie-terminal fraction / mean word depth	+0.856

The Lie-bridge-only fraction has zero variance on this ensemble and is removed before standardization. PCA on the seven retained diagnostics gives three components for both 90% and 95% of the observed variance. The leading axis is a reachability/depth scale; later axes separate repair, bridge-channel, and observable-rate effects. This is an empirical dimension estimate for

the tested ensemble, not a proof of a minimal, complete, or orthogonal invariant basis. It is also ensemble-composition dependent: changing the relative mix of GridWorld, SIR, random-graph, and random-skew configurations can change the explained-variance percentages.

Trajectory diagnostics are reported separately. The present three trajectory controls are insufficient for a trajectory-invariant PCA, so no independence claim is made for oscillation, plateau, or persistence coordinates.

Example 3.1.

Species / probe	Informal signature
Rubik spectral deformation	finite, collision-like, smooth-branch candidate, higher-order endpoint
Rubik Σ_{comm} slice	sparse, high-codimension, persistent commutativity anchor, not an accessibility map
Rubik accessibility repair	repairable, finite, incidence/support-driven
Absorbing Markov chain	terminal-side, absorbing, monotone/frozen, unrepaired
Yang-like state mixing	monotone degeneration or plateau degeneration, comparison branch
Transformer activation SOF	sparse repair, activation-induced, diagnostic token-sector influence
Barrier-option SOF	stochastic stopping boundary, cross-barrier support, hitting-time diagnostic

4 Observable Wall Taxonomy

Paper XI organizes signatures into a taxonomy before attempting any local smooth model.

4.1 Taxonomy Classes

Class	Name	Diagnostic pattern	Typical examples
A	Collision walls	spectral or discriminant branches collide or merge	Rubik spectral deformation, smooth spectral probes, NCG spectral-block diagnostics
B	Repair walls	an inaccessible or frozen channel becomes accessible at a higher observable layer	Rubik repair, quantum D -repair, transformer sparse repair
C	Terminal-side walls	trajectories enter a temporarily terminal, dead, or absorbing side of a deformation; record whether repair occurs	absorbing Markov chains, pre-repair quantum or MoE diagnostics, barrier boundaries
D	Plateau walls	observables remain flat, delayed, oscillatory, or degenerate over intervals	Yang-like degeneration, training plateaus, grokking-style delayed rates
E	Nonsmooth or discrete walls	wall is induced by a discrete jump, rank selection, or piecewise-smooth kink	graph rewiring, ReLU kinks, Top-k activation selection
F	Bridge or incidence walls	bridge products, rank incidence, or algebraic association controls wall behavior	Rubik Type III/IV mechanisms, incidence products, bridge-level audits

4.2 Observable Normal Forms

Before invoking smooth singularity theory, wall signatures can be grouped into SOF-native observable normal forms. These are descriptive normal forms for observable signatures, not theorem-level ADE classes.

Observable Wall Taxonomy

Six registered signature classes precede any smooth singularity model

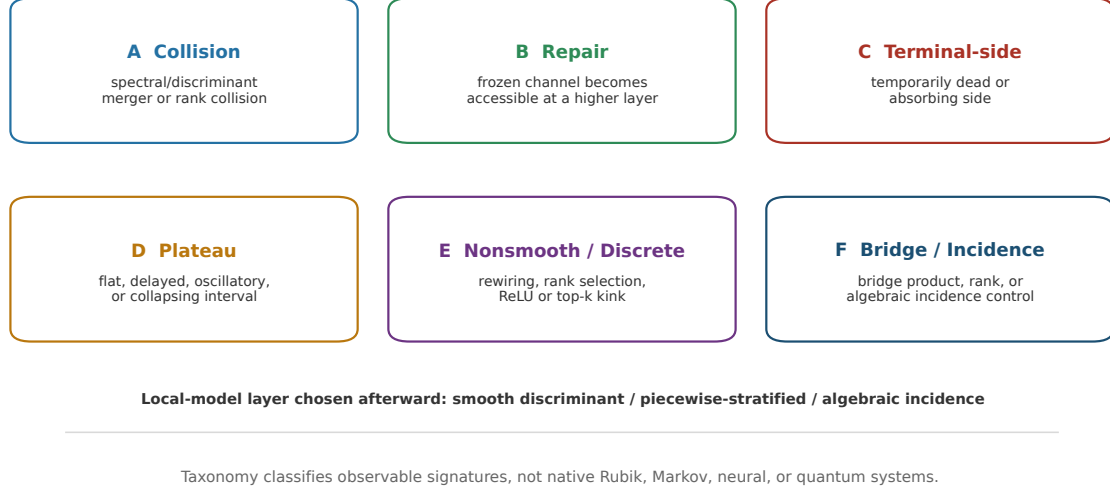


Figure 3. Observable Wall Taxonomy. The six registered signature classes are collision, repair, terminal-side, plateau, nonsmooth/discrete, and bridge/incidence walls. The classes precede smooth, stratified, or algebraic local-model choices and do not classify native systems directly.

Normal form	Signature pattern	Typical diagnostic
NF-1: single rank collision	one spectral/rank component collides or merges	one A_1 -like gap closure or one rank jump
NF-2: bridge creation	a previously frozen pair becomes accessible at R_2 or D	positive D_{repaired} or bridge support
NF-3: oscillatory repair	repair or plateau observable alternates before stabilizing	nonzero oscillation index O_W
NF-4: plateau collapse	plateau interval degenerates, flattens, or terminates	monotone degeneration or endpoint collapse
NF-5: terminal absorption	a state, sector, or channel enters an absorbing/frozen terminal regime	frozen pairs with no repair
NF-6: bridge/incidence association	a bridge product, image-kernel relation, or algebraic incidence controls the wall	Rubik Type III/IV and rank-protected bridge audits

Observable wall records are first sorted by five axes.

4.3 Smoothness Type

Type	Description	Classification status
Smooth discriminant	$O(t)$ is smooth and Σ_O is a discriminant pullback	ADE may apply locally
Piecewise-smooth wall	$O(t)$ is continuous but not smooth across strata	needs stratified or piecewise theory
Discontinuous jump	$O(t)$ changes by a discrete event or rank jump	outside classical ADE
Degenerate/nonresponsive trajectory	is flat, decreasing, zero, or not a response curve	not a wall class by itself
Incidence mechanism	wall is defined by algebraic product/rank incidence	algebraic-geometry classification needed

4.4 Observable Layer

Layer	Typical observable	Typical wall type
Spectral	eigenvalues, collision quotients	spectral collision/discriminant
Accessibility	$R_1/R_2/D$, $\mathcal{J}_{\text{acc}}$	rank/support/accessibility discriminant
Filtration	plateau functions $P_d(t)$	degeneration or plateau transition
Markov	communicating-class or transition support	absorbing/frozen boundary
Graph	Laplacian gap, adjacency transport	discrete rewiring or spectral sensitivity
Quantum	gate-log accessibility, controllability	channel/accessibility transition
Neural-network	activation sectors, $K_0/K_1/K_2$	training-coupled or activation-induced wall

4.5 Deformation Geometry

The same observable ladder can produce different wall behavior depending on what changes: generator weights, state mixing, gate interpolation, edge rewiring, training dynamics, or rate-matrix perturbation.

Therefore wall classification must be attached to a deformation geometry, not only to the SOF object.

5 Smooth Wall Models

Smooth wall models are the final layer, not the starting point. Arnold ADE classification is a plausible local model only for sufficiently smooth discriminant branches. A typical admissible situation would have:

$$O : T \rightarrow E, \quad \Sigma_O = O^{-1}(\Delta),$$

with O smooth and Δ a stable local singularity.

In this restricted setting, the local wall may fall into an A_k , D_k , or $E_{6,7,8}$ class.

This paper does not assert such a classification globally. In particular:

1. graph rewiring is discrete;
2. Top-k and step activations are discontinuous;
3. ReLU-type walls are piecewise-smooth;

4. accessibility incidence can be algebraic rather than ADE-like;
5. flat or decreasing trajectories are not rate-response walls.

ADE is therefore a local smooth-branch model, not the SOF classification framework.

The branch-aware A_n adjacency falsification audit is recorded in Appendix A. Its role in the main text is only to prevent a false promotion of sorted-index pair-gap responses into an ADE adjacency claim.

6 Cross-Species Audit

The computational audit combines cross-species wall diagnostics, smooth spectral pair-gap responses, record coverage, snapshot redundancy, branch-aware adjacency falsification, sampled trajectory events, and two registry-handoff diagnostics. The artifact map is collected in Appendix C. The discriminant-bifurcation scan is an auxiliary 2D-slice diagnostic; it is not a full map of Σ_{access} .

6.1 Rubik Spectral Snapshot

The Rubik A_{18} audit reports six spectral layers with eigenvalues

$$1, 8/9, 7/9, 2/3, 5/9, 1/3,$$

and k -set $\{0, 1, 2, 3, 4, 6\}$. This is a fixed spectral snapshot. It is not, by itself, a moving wall.

6.2 Rubik Spectral Pair-Gap Local-Model Audit

The spectral-deformation audit varies one QT generator weight and tracks the eigenvalues of a fixed separating Hermitian probe

$$M(\alpha) = QT(\alpha) + \beta HT(\alpha), \quad \beta = 0.314159.$$

The audit detects 16 pairwise adjacent-gap closures at the canonical endpoint. Each closure is recorded as an A_1 /fold candidate in the smooth spectral branch. The endpoint itself is higher order: the regular QT/HT sector point is a simultaneous multi-branch collapse rather than a single isolated fold.

The same audit also runs a two-weight diagonal search and records simultaneous pair-gap responses. The observed count ranges from one to three across tested sampling resolutions, so the integer is not promoted to a stable invariant. The two gaps need not share an eigenvalue branch, and the audit does not perform eigenbranch continuation. These observations therefore remain pair-gap response diagnostics; they are not identified as A_2 /cusp unfoldings and do not prove an ADE classification theorem for SOF walls.

6.3 Rubik 2D Σ_{comm} Slice

The discriminant-bifurcation audit scans a two-dimensional plane of QT generator weights and tests the Frobenius norm of

$$[QT(w), HT(w)].$$

On the 20×20 slice varying two QT generator weights, the audit reports one hit on the chosen 20×20 evaluation grid. The single commutative cell has 9 spectral points, and the audit

records no spectral point-count bifurcations inside the commutative cells. This is not a proof of the codimension-11 commutativity theorem from Paper VI and not a map of Σ_{access} . Its role in Paper XI is narrower: it gives a sampling anchor showing that arbitrary two-dimensional generator-weight slices meet the high-codimension commutativity locus sparsely. Spectral and accessibility wall taxonomy should therefore be computed on normal spectral charts or on deformation models chosen for that purpose, not inferred from a generic 2D scan alone.

6.4 Quantum Accessibility Contrast

The quantum audit compares Pauli, Clifford+CNOT, and Universal+CNOT gate-sector SOFs. Clifford+CNOT and Universal+CNOT show D -repair, while the Pauli control does not. This is an accessibility contrast outside Rubik, not a claim that quantum systems obey Rubik wall theory.

A separate repair-persistence audit interpolates CNOT strength in matrix space from a product endpoint to the Clifford+CNOT gate-log SOF. On the 21-point grid $0, 0.05, \dots, 1$, D_{repaired} is zero through strength 0.50 and jumps to 6 at strength 0.55. The repair then persists through strength 1.00 with no reversal. Thus the initial repair-persistence signature is threshold 0.55, $p_W = 0.45$ over the unit interpolation range, and post-activation stability 100%. In grid-count terms this corresponds to 10 active samples out of 21; p_W records the continuous parameter range, not the sample fraction.

6.5 Markov Frozen-Pair Contrast

The Markov audit separates complete/strongly connected examples from an absorbing-state example. The absorbing case has true frozen pairs. This is a communicating-class or absorbing-boundary diagnostic, not an ADE classification.

6.6 Graph Spectral-Gap Sensitivity

The graph audit records Laplacian-gap changes under edge removal. This is a discrete perturbation diagnostic. It is not a smooth codimension-one wall without a separate continuous edge-weight model and discriminant calculation.

6.7 Piecewise-Smooth Activation Wall Boundary

The activation-wall audit compares ReLU, GeLU, and Top-k sector diagnostics. In the default finite test, the ReLU sector diagnostic has distinct left and right slopes, producing a piecewise-smooth kink. GeLU is smooth in the same test and does not define a kink wall. Top-k is a rank-selection diagnostic and is treated as a discontinuous or discrete wall record.

This audit supports the negative boundary for ADE usage: activation-induced SOF walls may require piecewise-smooth, stratified, or discrete wall theory rather than classical smooth ADE singularity theory.

6.8 Barrier-Option Stopping Boundary

The barrier-option SOF uses a finite log-price grid with below-barrier and above-barrier sectors. The default audit reports cross-barrier generator support with $R_1 = 75.0\%$, $R_2 = 0.0\%$, no D -repair, and mean first-hit time 6.5915 from the chosen initial grid point to the barrier sector. This is a stochastic stopping-region wall record. The first-hitting time is a native stochastic diagnostic, not the SOF depth D , and the entry is not an option-pricing theorem.

6.9 Transformer Activation-Sector Diagnostic

The transformer diagnostic constructs a token-space SOF from activation-count clusters in a tiny transformer-like block. Attention top-k and FFN activation-similarity operators give $R_1 = 58.3\%$, $R_2 = 66.7\%$, $D_{\text{repaired}} = 2$, and $D_{\text{max}} = 2$ in the default audit. This is an activation-induced diagnostic wall record: it shows how token-sector decompositions can expose cross-sector influence, but it is not a theorem about all transformers or LLM explainability.

7 Boundary Audit Notes

Two boundary audits are deferred to the appendices. Appendix A records the branch-aware adjacency falsification audit for Rubik spectral pair responses. Appendix B records the observable-status trajectory audit. Their shared main-text role is cautionary: sorted gaps should not be promoted to ADE adjacency without branch continuation, and static inaccessible pairs should not be relabeled as walls without a change along a declared deformation path.

8 Working Taxonomy Table

The table below is a first finite approximation to Spec_W across the registered examples. Each row records one wall signature or wall-spectrum entry; a single species may contribute more than one row in future registry versions.

Species / probe	Class	Wall signature	Smooth-model status	Claim status
Rubik spectral snapshot	A pre-wall reference	fixed finite spectral arrangement	not a moving wall by itself	exact snapshot
Rubik spectral deformation	A collision wall	higher-order endpoint; 16 A_1 -type pairwise closures and simultaneous pair-gap responses	branch-aware audit finds no $A_2 \rightarrow A_1 + A_1$ split candidate	smooth-branch diagnostic, not full classification
Rubik Σ_{comm} 2D slice	A/B boundary anchor	sparse high-codimension intersection; one hit on the chosen 20×20 evaluation grid	sampling anchor; codimension-11 belongs to the Paper VI tangent theorem	boundary diagnostic, not a full discriminant map
Accessibility $R_1/R_2/D$	B repair wall	rank/support/accessibility jump; repairable or unrepaired depending on species	possible smooth branch on Σ_{spec}	conditional
Rubik Type III/IV bridge mechanisms	F bridge/incidence wall	bridge products, cancellation, and rank-incidence structure; naturally carried by Rubik	algebraic/incidence normal form, not ADE by default	Paper V–VII and Paper X registry evidence
Quantum accessibility	B repair wall	accessibility-channel contrast; D -repair appears for entangling gate sets	not classified	diagnostic evidence
Markov absorbing boundary	C terminal-side wall	absorbing, frozen, terminal-side, unrepaired	not ADE by default	diagnostic evidence
Graph edge rewiring	E discrete wall	discrete spectral-gap sensitivity under edge removal	outside smooth ADE	diagnostic evidence
Barrier-option stopping sector	C/D stochastic boundary	cross-barrier support with first-hitting diagnostic; hitting time is not SOF depth	not ADE by default	registry diagnostic evidence
NN training dynamics	D plateau/rate wall	delayed or ordered proxy response; rate hierarchy is not a wall class by itself	smooth only with explicit training model	Paper IX/X evidence
ReLU / Top-k activations	E nonsmooth wall	piecewise kink or rank-selection jump	requires nonsmooth or stratified extension	boundary diagnostic evidence
Transformer activation sectors	B/E activation repair wall	sparse token-sector repair; $D_{\text{repaired}} = 2$ in default audit	activation-induced; requires diagnostic/stratified theory	diagnostic handoff evidence
Yang-like state mixing	D plateau wall	monotone degeneration or plateau degeneration	smooth candidate only with explicit model	comparison branch

9 What This Paper Does Not Claim

Paper XI does not claim:

1. a complete classification of all SOF trajectories;

2. that Arnold ADE applies to all SOF branches;
3. that graph rewiring is a smooth codimension-one wall;
4. that activation-induced walls fit into ADE without extension;
5. that the mechanism-separation principle of Paper X is a universal theorem;
6. that rate hierarchy alone determines wall class;
7. that Paper XI supplies general transformation laws for observables across walls;
8. that the wall assignment is full, faithful, or invariant under all weak SOF comparisons;
9. that the listed observable invariants form a complete invariant system;
10. that observable normal forms are already classification theorems;
11. that the tested Rubik spectral slices exhibit A_n adjacency.

The stable claim is narrower:

SOF wall behavior is first organized by observable wall records, wall signatures, wall spectra, and computable observable invariants. The Observable Wall Taxonomy groups those signatures into collision, repair, terminal-side, plateau, nonsmooth/discrete, and bridge/incidence classes. Observable normal forms organize the registered signatures before any smooth singularity model is invoked. ADE-type local modeling is one candidate subtheory for sufficiently smooth wall records, not the whole theory. In this sense, Paper XI is the observable feature extractor between the Paper X Registry and later SOF diagnostic workflows.

10 Conclusion

Paper XI turns SOF wall phenomena into a record-level classification problem. Its basic output is not a universal wall-crossing law, but a sequence of observable records, signatures, wall spectra, and diagnostic invariants that can be compared across registered SOF species. Smooth ADE-style normal forms remain candidate local models only on sufficiently smooth discriminant branches; the taxonomy itself also admits repair, terminal-side, plateau/rate, non-smooth/discrete, and bridge/incidence records. This paper classifies observable wall records rather than wall evolution laws.

11 Outlook

Paper XI supplies a record-level taxonomy, not a classification theorem. The next stage is to decide which parts of the taxonomy are invariant under comparison of SOF presentations, which parts depend on the chosen ladder or sampling path, and which parts admit local geometric normal forms.

The remaining classification program has five parts:

1. formalize comparison and equivariance conditions for $\text{WallAssign}(\mathcal{F}, \mathcal{L}, \Gamma)$: defining compatible comparison data is straightforward, but proving invariance under weak SOF comparisons, path reparametrizations, and sampling refinements is a separate theorem-level problem;

2. build continuous graph and Markov deformation models where smooth discriminants can be computed;
3. compute local codimensions for accessibility discriminants and higher-order spectral endpoints;
4. separate incidence varieties from smooth singularity classes;
5. develop or import piecewise-smooth and stratified wall theory for activation-induced and rank-selection SOFs.

Only after these ingredients and additional registered examples are available can the taxonomy be promoted toward classification theorems.

Appendix A Branch-Aware Adjacency Falsification Audit

Simultaneous small gaps in a sorted full-spectrum list are not enough to establish A_n adjacency. Such gaps may belong to different invariant blocks, may exchange order, or may be symmetry-protected degeneracies rather than isolated collision branches.

The branch-aware audit therefore separates the Rubik representation into its *cp*, *ep*, *co*, and *eo* blocks and continues eigenbranches by maximizing eigenvector overlap between adjacent parameter samples. A candidate $A_2 \rightarrow A_1 + A_1$ split requires two isolated pair-gap minima at separated parameter values whose union involves all three branches. Persistent degeneracies, endpoint-only closures, and cross-block equalities are rejected.

Block	Dimension	Endpoint cluster sizes	Tested triples	Split candidates
<i>cp</i>	64	8, 24, 24, 8	0	0
<i>ep</i>	144	12, 36, 24, 36, 36	0	0
<i>co</i>	8	3, 3	2	0
<i>eo</i>	12	3, 3, 3	0	0

On the tested diagonal and asymmetric slices, no branch-aware $A_2 \rightarrow A_1 + A_1$ split candidate is detected. The earlier sorted-index pair-gap responses remain useful spectral diagnostics, but they do not support an A_n adjacency claim. Even a future positive candidate would still require local normal-form and versality analysis before becoming an ADE statement.

Appendix B Observable-Status Wall Trajectories

Given audit snapshots at parameters $t_0, \dots, t_K$, assign each ordered pair one observable status:

$$S_{ij}(t) \in \{\text{direct, word bridge, Lie bridge, deeper, terminal}\}.$$

A sampled wall event occurs only when

$$S_{ij}(t_{k-1}) \neq S_{ij}(t_k).$$

The trajectory implementation records every adjacent-step change, including repeated changes of the same pair. Events are labeled as repair, terminalization, support gain, support loss, or layer change. Stable terminal pairs remain trajectory metadata and are not counted as walls.

Three controlled paths are implemented:

Control	Steps	Ordered pairs	Events	Changed pairs	Event steps
GridWorld obstacle path	3	600	190	140	1, 2
SIR β sweep	21	6	4	4	1
Graph edge-weight endpoint	11	12	6	6	10

The SIR result isolates all changes at the first sampled step $\beta = 0 \rightarrow 0.025$. The graph control is constant for $t < 1$ and changes only when the weakened edges vanish at $t = 1$. GridWorld records two discrete topology replacements and therefore has events at both steps.

These are sampled-path wall records. They do not establish ambient codimension, continuous-time wall flow, or a bifurcation theorem. Their role is narrower: they provide a correct dynamic record layer between Paper IX deformations and the Paper XI taxonomy without relabeling every frozen pair as a wall.

Appendix C Computational Artifacts

The following repository artifacts provide the reproducible support layer for the audits reported in this paper. The main text refers to audit roles rather than file paths; this appendix records the exact implementation and generated tables. Unless a prefix is shown, files are located under `experiments/paper11/`.

Artifact	Role in Paper XI	File
C1	cross-species wall diagnostics	<code>cross_species_wall_audit.py</code>
C2	smooth spectral pair-gap local-model audit	<code>spectral_ade_collision.py</code>
C3	auxiliary 2D commutativity-discriminant slice	<code>discriminant_bifurcation_map.py</code>
C4	wall-density taxonomy sample	<code>wall_density_registry.py</code>
C5	wall-record coverage census	<code>wall_record_census.py</code>
C6	generated wall-record census tables	<code>results/wall_record_census.md, .json</code>
C7	definition-compatible redundancy audit	<code>invariant_redundancy.py</code>
C8	branch-aware adjacency falsification	<code>an_adjacency.py</code>
C9	observable-status trajectory records	<code>wall_trajectory.py</code>
C10	CNOT-strength repair persistence	<code>repair_persistence_quantum.py</code>
C11	piecewise-smooth activation boundary	<code>piecewise_smooth_activation_wall.py</code>
C12	barrier-option registry handoff	<code>paper10/barrier_option_sof.py</code>
C13	transformer activation-sector handoff	<code>paper12/transformer_activation_sof.py</code>

Bibliographic Notes

Program lineage. Paper XI depends on Papers VII–X. Paper VII supplies the generic completion and repair boundary [1]; Paper VIII supplies the SOF object layer [3]; Paper IX supplies observable trajectories and wall pullbacks [2]; Paper X supplies the Universal Observable Pipeline and five-layer SOF Registry [4].

Smooth singularities. Arnold’s catastrophe and ADE singularity theory is relevant only to the smooth-discriminant branch. Useful background includes Arnold’s *Catastrophe Theory*,

the work of Arnold, Gusein-Zade, and Varchenko in *Singularities of Differentiable Maps*, and Golubitsky and Guillemin’s *Stable Mappings and Their Singularities*.

Stratified and nonsmooth walls. Piecewise smooth, discontinuous, and multi-stratum SOF walls require different tools. Relevant background includes Whitney and Thom–Mather stratification, Goresky and MacPherson’s *Stratified Morse Theory*, the work on piecewise smooth dynamical systems by di Bernardo, Budd, Champneys, and Kowalczyk, and Filippov’s discontinuous differential equations.

Spectral and finite-state diagnostics. Graph and Markov wall records are compared against spectral perturbation theory, spectral graph theory, and Markov chain perturbation and mixing theory before any singularity classification is claimed.

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